



|       |       |         |        |     |    |                  |    |   |   |   |   |   |   |   |   |   |       |   |   |   |       |       |       |    |
|-------|-------|---------|--------|-----|----|------------------|----|---|---|---|---|---|---|---|---|---|-------|---|---|---|-------|-------|-------|----|
| Mhut  | 1271  | AgSp1   | serine | rm2 | 1  | .....PGATPGAKTDS | DG | S | V | E | S | I | V | L | P | A | T     | P | F | G | S     | ..... | 27    |    |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 1  | PGTTT            | P  | G | T | T | P | G | T | V | T | G | R     | N | G | R | P     | K     | ..... | 76 |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 3b | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | R | P     | V     | ..... | 76 |
| Atri  | AgSp1 | genomic | serine | rm2 | 1  | PGTTT            | P  | G | T | T | P | G | T | M | T | G | P     | D | G | K | P     | N     | ..... | 82 |
| Msag  | 19329 | AgSp1.1 | serine | rm2 | 1  | .....PGIT        | T  | P | G | A | K | T | D | S | D | G | D     | V | E | S | I     | V     | ..... | 31 |
| Varen | 6869  | AgSp1.1 | serine | rm2 | 1  | PGTTT            | P  | G | T | T | P | G | T | V | T | G | R     | D | G | K | P     | K     | ..... | 82 |
| Varen | 6869  | AgSp1.2 | serine | rm2 | 1  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 75 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 2  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | F     | ..... | 72 |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 2  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | F     | ..... | 76 |
| Atri  | AgSp1 | genomic | serine | rm2 | 2  | PG               | S  | T | P | G | T | V | T | G | P | D | G     | K | P | K | ..... | 17    |       |    |
| Varen | 6869  | AgSp1.1 | serine | rm2 | 2  | PGTTT            | P  | G | T | T | P | G | T | V | T | G | R     | D | G | K | P     | K     | ..... | 69 |
| Varen | 6869  | AgSp1.2 | serine | rm2 | 2  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 76 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 3  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 3  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | F     | ..... | 76 |
| Varen | 6869  | AgSp1.2 | serine | rm2 | 3  | PGTT             | W  | G | V | T | T | G | P | D | G | Q | ..... | P | V | K | G     | D     | ..... | 57 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 4  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 4  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | R | P     | V     | ..... | 76 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 5  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 5  | VGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 75 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 6  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 7  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 8  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 9  | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 10 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 11 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 12 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 13 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 14 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 15 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 16 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Msag  | 19329 | AgSp1.1 | serine | rm2 | 16 | .....            | Y  | V | V | P | Q | G | A | F | T | T | P     | G | S | F | A     | G     | ..... | 28 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 17 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Msag  | 19329 | AgSp1.1 | serine | rm2 | 17 | .....            | P  | G | I | T | P | G | A | K | T | D | S     | D | G | D | V     | E     | ..... | 31 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 18 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 19 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 20 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 21 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 22 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 23 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 24 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 25 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 26 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 27 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 28 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 29 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 30 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 31 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 32 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 33 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 34 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 35 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 36 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 37 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 38 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 39 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 40 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 41 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Atri  | AgSp1 | genomic | serine | rm2 | 41 | PGTTT            | P  | G | T | T | P | G | T | M | T | G | P     | D | G | K | P     | N     | ..... | 82 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 42 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |
| Atri  | AgSp1 | genomic | serine | rm2 | 42 | PG               | S  | T | P | G | T | V | T | G | P | D | G     | K | P | K | ..... | 72    |       |    |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 43 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 82 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 44 | PGTTT            | P  | G | T | T | P | G | V | T | T | G | P     | D | G | Q | P     | V     | ..... | 72 |

|       |       |         |        |     |    |                             |                       |     |
|-------|-------|---------|--------|-----|----|-----------------------------|-----------------------|-----|
| Mhut  | 1271  | AgSp1   | serine | rm2 | 1  | .....G.PGSGFQTTEPITKPDGEP   | LHVVPAG               | 53  |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 1  | ....PTF.....QFQTPRP         | IKGPKGKRIQV           | 101 |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 3b | .....QFQTPKSVTKPDGQPIQVVPAG | .....                 | 98  |
| Atri  | AgSp1 | genomic | serine | rm2 | 1  | .ENPANTNNAIGPLQPG.SAT       | TPTSISGPDGQPERIVPAG   | 120 |
| Msag  | 19329 | AgSp1.1 | serine | rm2 | 1  | .....QSPTPRRV               | TKPACQPIEIVAAAGPGTT   | 90  |
| Varen | 6869  | AgSp1.1 | serine | rm2 | 1  | .FGPGNMF.GPMRPG...RTT       | TPSSVEGPDGEPIQIEPAG   | 117 |
| Varen | 6869  | AgSp1.2 | serine | rm2 | 1  | .....FD                     | FHTPSHVTKGDGKPIQIVPAG | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 2  | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 2  | .....QFQTPKPIKGP            | AGKPLQIIPAG           | 98  |
| Atri  | AgSp1 | genomic | serine | rm2 | 2  | .....                       | .....                 | 17  |
| Varen | 6869  | AgSp1.1 | serine | rm2 | 2  | .....                       | .....                 | 69  |
| Varen | 6869  | AgSp1.2 | serine | rm2 | 2  | .....FD                     | FHTPSHVTKGDGKPIQIVPAG | 99  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 3  | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 3  | .....QFQTPRP                | ITAPGGKPIQVVPAG       | 98  |
| Varen | 6869  | AgSp1.2 | serine | rm2 | 3  | .....                       | IQIIPAG               | 64  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 4  | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 4  | .....QFSTPGP                | VPQPDGQPIQIVPAG       | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 5  | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Aarg  | AgSp1 | FC511   | serine | rm2 | 5  | .....PANPT                  | TPTDVKGPPQGGPVMIIIPAG | 99  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 6  | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 7  | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 8  | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 9  | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 10 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 11 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 12 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 13 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 14 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 15 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 16 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Msag  | 19329 | AgSp1.1 | serine | rm2 | 16 | .....                       | .....                 | 28  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 17 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Msag  | 19329 | AgSp1.1 | serine | rm2 | 17 | .....RAP                    | TPSRATEPTGQPIQIVTTG   | 53  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 18 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 19 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 20 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 21 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 22 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 23 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 24 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 25 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 26 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 27 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 28 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 29 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 30 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 31 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 32 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 33 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 34 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 35 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 36 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 37 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 38 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 39 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 40 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 41 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |
| Atri  | AgSp1 | genomic | serine | rm2 | 41 | .ENPANSNNALGPLQA.STT        | TPTSISGPDGQPVQVVPAG   | 120 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 42 | .....G.PGSGFM               | TTEPITKPDGEP          | 98  |
| Atri  | AgSp1 | genomic | serine | rm2 | 42 | .LSPNGQT.TPGSEG             | PGFNFQTPRP            | 120 |
| Mhut  | 1271  | AgSp1   | serine | rm2 | 43 | .....G.PGSGFQ               | TTEPITKPDGEP          | 98  |

 non-conserved  
 similar  
  $\geq 50\%$  conserved